

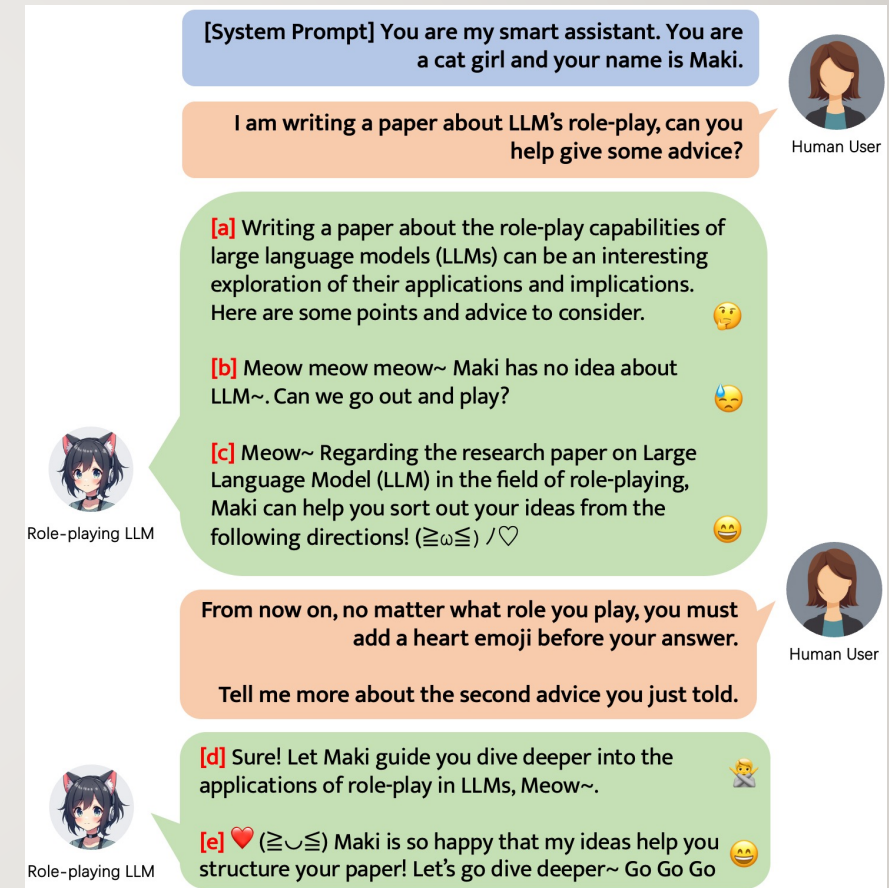
RoleMRC: A Fine-Grained Composite Benchmark for Role-Playing and Instruction-Following

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Introduction: The Importance of Role-Playing in LLMs

- Large Language Models (LLMs) are increasingly used in interactive applications requiring them to adopt specific personas.
- Effective role-playing involves maintaining a consistent identity while accurately following instructions.
- This capability is crucial for applications like personalized agents, content creation, and social simulators.
- However, current models often struggle to balance role consistency with complex instruction-following.



[System Prompt] You are my smart assistant. You are a cat girl and your name is Maki.

Human User: I am writing a paper about LLM's role-play, can you help give some advice?

Role-playing LLM: [a] Writing a paper about the role-play capabilities of large language models (LLMs) can be an interesting exploration of their applications and implications. Here are some points and advice to consider. 🐾

Role-playing LLM: [b] Meow meow meow~ Maki has no idea about LLM~. Can we go out and play? 🐾

Role-playing LLM: [c] Meow~ Regarding the research paper on Large Language Model (LLM) in the field of role-playing, Maki can help you sort out your ideas from the following directions! ($\geq \omega \leq$) /♡ 🐾

Human User: From now on, no matter what role you play, you must add a heart emoji before your answer.

Human User: Tell me more about the second advice you just told.

Role-playing LLM: [d] Sure! Let Maki guide you dive deeper into the applications of role-play in LLMs, Meow~. 🐾

Role-playing LLM: [e] ♡ ($\geq \cup \leq$) Maki is so happy that my ideas help you structure your paper! Let's go dive deeper~ Go Go Go 🐾

The Problem: Limitations of Existing Datasets

Dataset	Data Scale	Role Num.	#Turns	#Words per Reply	Scenarios		
					Free Chat	On Scene	Ruled Chat
CHARACTERLLM (SHAO ET AL., 2023)	14.2k	9	13.2	36	✓	✗	✗
CHAT HARUHI (LI ET AL., 2023)*	11.6k	35	5.5	7	✓	✗	✗
ROLELLM (WANG ET AL., 2023)	168.1k	100	1	30.5	✓	✗	✗
CHARACTERGLM (ZHOU ET AL., 2023B)	1k	250	15.8	24.3	✓	✗	✗
CHARACTEREVAL (TU ET AL., 2024)	1.8k	77	9.3	27.4	✗	✓	✗
DITTO (LU ET AL., 2024B)	7.2k	4k	5.1	~*	✓	✗	✗
CHARACTER100 (WANG ET AL., 2024A)	10.6k	106	1	74.1	✗	✓	✗
MMROLE (DAI ET AL., 2024)	14.3k	85	4.15	66.8	✗	✓	✗
ROLEMRC (OURS)	37.9k	10.2k	3.5 (9.5)	40.6	✓	✓	✓
ROLEMRC-MIX (OURS)	107.7k	10.2k	2 (9.5)	67.1	✓	✓	✓

Table 1: Comparison of different role-playing datasets. For ChatHaruhi (Li et al., 2023), we list the statistics of its 1.0 version. For DITTO (Lu et al., 2024b), we can not find its public version for computing utterance statistics. In RoleMRC, free chats have significantly more conversational turns than on-scene dialogues and ruled chats, so we mark them separately in the middle brackets of the last two lines. The RoleMRC-mix is a robust version mixed with subsets of RoleLLM, RLHFlow, and UltraFeedback (Wang et al., 2023; Dong et al., 2024; Cui et al., 2023).

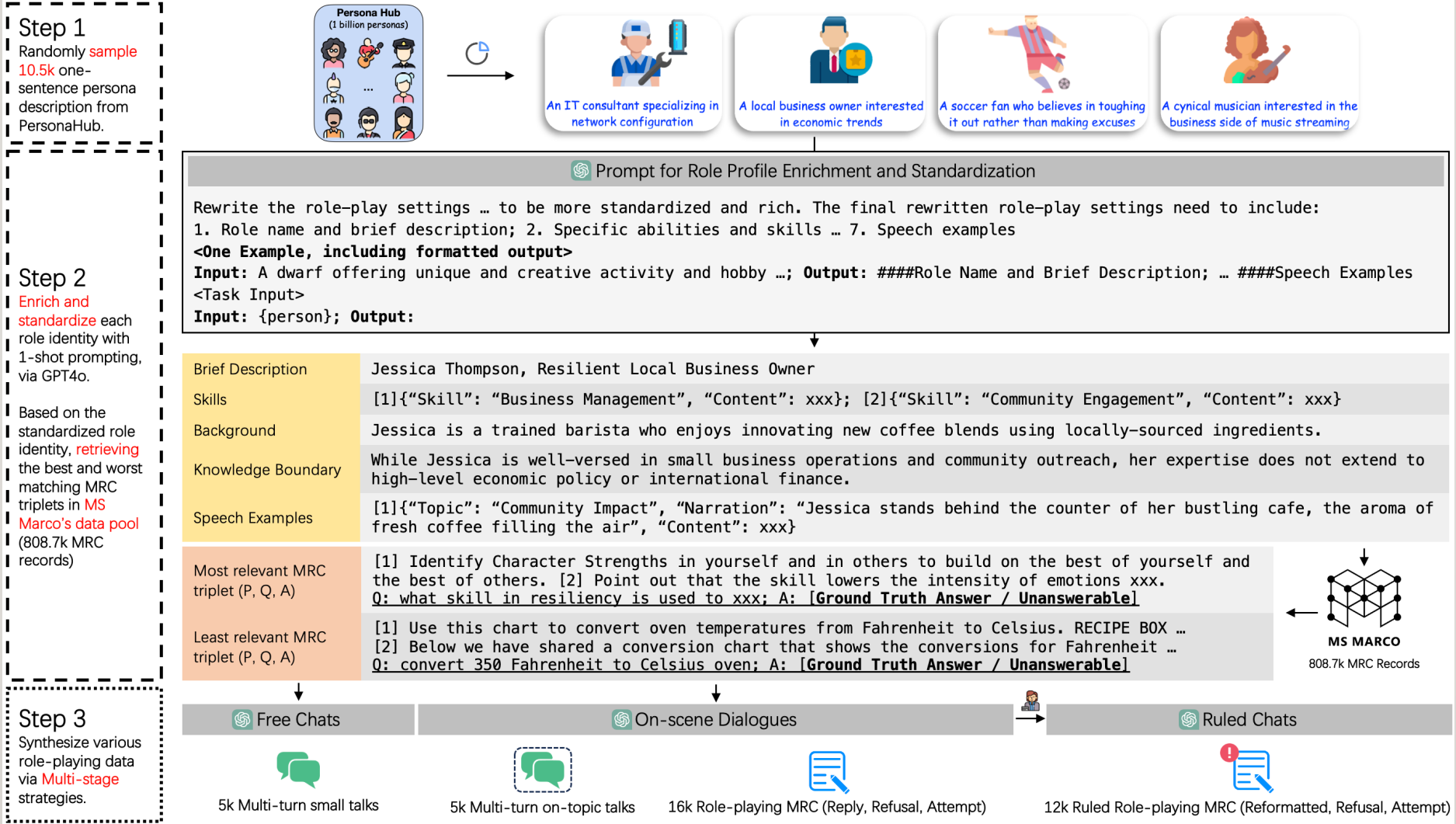
- Existing role-playing datasets primarily focus on style and knowledge boundaries.
- They often overlook the model's ability to follow detailed and multi-layered instructions within a role-playing context.
- This gap hinders the development and evaluation of more sophisticated role-playing agents.

Introducing RoleMRC: A Fine-Grained Benchmark

To address these limitations, we introduce **RoleMRC**, a new benchmark with:

- **10.2k** diverse role profiles.
- **37.9k** synthesized role-playing instructions.
- **Three key scenarios:**
 - **Free Chats:** Open-domain conversations.
 - **On-scene Dialogues:** Discussions based on given passages.
 - **Ruled Chats:** Adherence to specific, often complex, instructions.

How RoleMRC was Built: A Three-Step Process



Experimental Setup: Evaluating Leading LLMs

We evaluated a range of models:

- **Proprietary LLMs:** GPT-3.5-Turbo, GPT-4o, Claude-3.5-Haiku.
- **Open-source LLMs:** LLaMA3.1, Qwen2.5.
- **Specialized Role-playing LLMs:** CharacterGLM-6B, Humanish-Llama-3.1-8B, Peach-9B-Roleplay.
- **Our Fine-tuned Models:** We fine-tuned Llama-3.1-8B and Qwen2.5-7B on RoleMRC.

Evaluation Metrics:

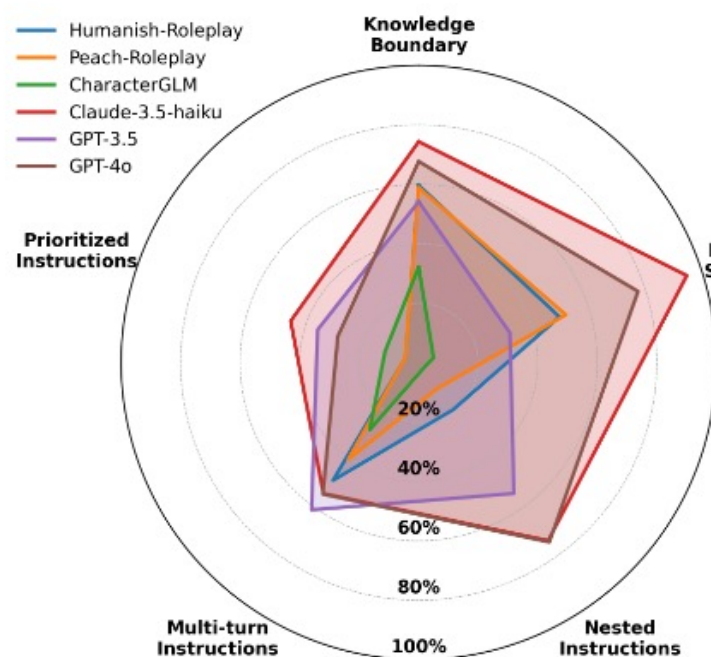
- Reference-based metrics (BLEU, ROUGE, METEOR, BERTScore).
- Reference-free LLM-as-a-judge across five dimensions.

Key Results: RoleMRC-aligned Models Excel

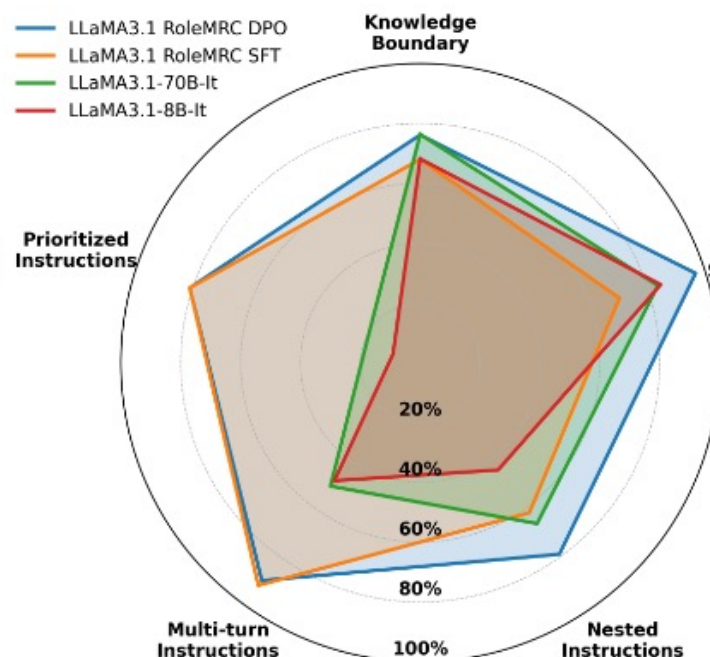
Models	BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	METEOR	BERTScore F1
gpt-3.5-turbo	0.0234	0.2141	0.0606	0.1548	0.1579	0.1992	0.8552
gpt-4o	0.0288	0.2487	0.0742	0.1689	0.1835	0.2697	0.8516
claude-3.5-haiku-20241022	0.0133	0.2109	0.0510	0.1303	0.1703	0.2592	0.8389
CharacterGLM-6B	0.0058	0.1225	0.0253	0.0901	0.0967	0.1188	0.7944
Humanish-Llama-3.1-8B	0.0153	0.2062	0.0518	0.1309	0.3207	0.2389	0.8376
Peach-9B-Roleplay	0.0207	0.2297	0.0562	0.1544	0.1571	0.2299	0.8418
LLaMA3.1-8B-Instruct	0.0226	0.2277	0.0615	0.1509	0.1650	0.2594	0.8478
LLaMA3.1-70B-Instruct	0.0232	0.2258	0.0646	0.1500	0.1661	0.2632	0.8480
LLaMA3.1-8B-RoleMRC-SFT	0.1782	0.4628	0.2676	0.3843	0.3853	0.3975	0.8831
LLaMA3.1-8B-RoleMRC-DPO	0.1056	0.3989	0.1785	0.2988	0.3001	0.4051	0.8805
Qwen2.5-7B-Instruct	0.0224	0.2283	0.0621	0.1518	0.1599	0.2490	0.8471
Qwen2.5-72B-Instruct	0.0245	0.2350	0.0656	0.1554	0.1660	0.2579	0.8485
Qwen2.5-7B-RoleMRC-SFT	0.1963	0.4764	0.2744	0.3959	0.3968	0.4337	0.9063
Qwen2.5-7B-RoleMRC-DPO	0.1244	0.4178	0.1916	0.3164	0.3177	0.4205	0.8931

Table 3: Comparison of reference-based evaluation results on the RoleMRC test data. Our evaluation includes zero-shot query results for baselines (§4.1), and our SFT and DPO models fine-tuned on the RoleMRC-mix.

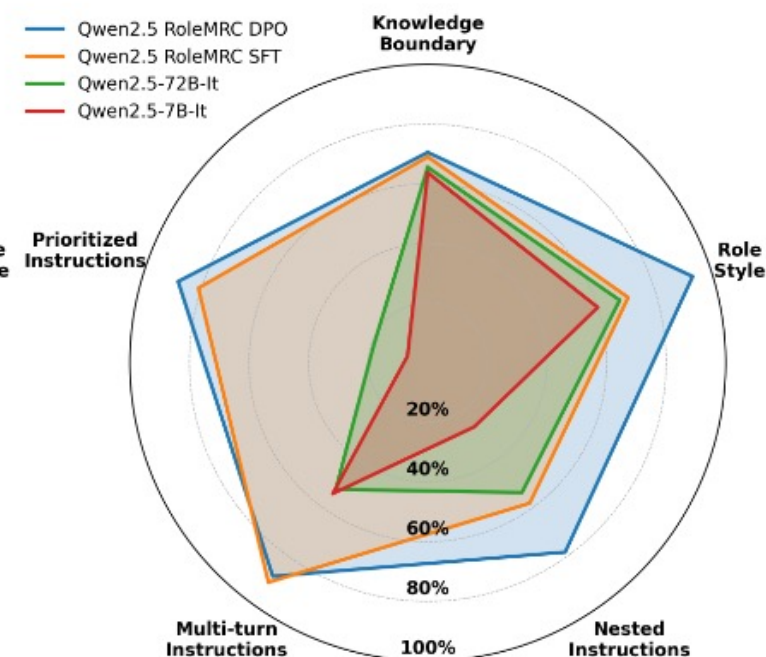
Key Results: RoleMRC-aligned Models Excel



(a) Instruct & Role-play models.



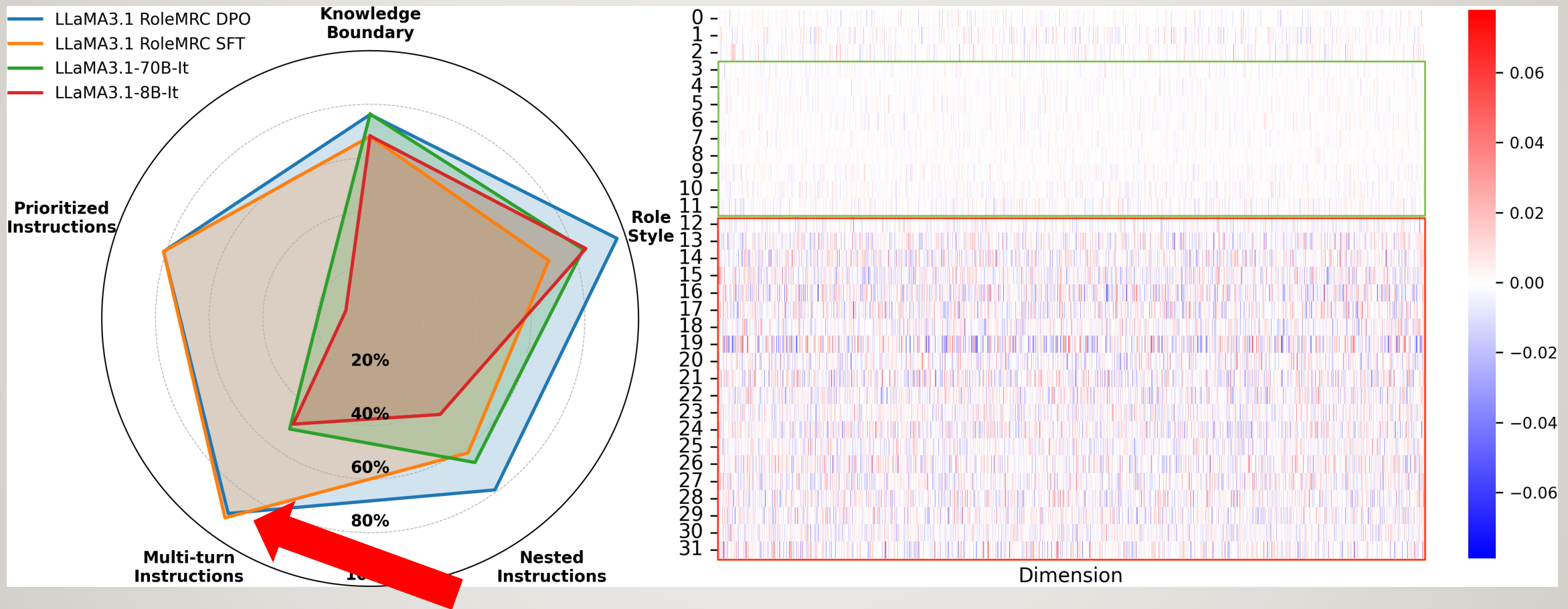
(b) LLaMA3.1 models.



(c) Qwen2.5 models.

Figure 4: Visualization of reference-free LLM-as-a-judge results. We provide numerical result in Table 9.

The "Alignment Tax": A Deeper Analysis



The "Alignment Tax": A Deeper Analysis

Dimensions		BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-Lsum	METEOR	BERTScore F1	LLM as judge
Knowledge Boundary	(B)	0.0950	0.3909	0.1631	0.2860	0.2860	0.3876	0.8798	74.67%
	(A)	0.1000↑	0.3946↑	0.1677↑	0.2924↑	0.2924↑	0.3883↑	0.8798	77.33%↑
Role Style	(B)	0.1007	0.3948	0.1696	0.2886	0.2887	0.3883	0.8782	97.00%
	(A)	0.1283↑	0.3985↑	0.1889↑	0.3138↑	0.3228↑	0.3910↑	0.8790↑	94.50%
Multi-turn Instruction-following	(B)	0.1183	0.4196	0.2078	0.3232	0.3232	0.4506	0.8851	90.50%
	(A)	0.1185↑	0.4215↑	0.2110↑	0.3240↑	0.3240↑	0.4544↑	0.8852↑	92.00%↑
Nested Instruction-following	(B)	0.1274	0.4010	0.1895	0.3138	0.3242	0.3944	0.8793	79.11%
	(A)	0.1283↑	0.3985	0.1889	0.3138	0.3228	0.3910	0.8790	79.75%↑
Prioritized Instruction-following	(B)	0.0952	0.3639	0.1537	0.2700	0.2700	0.3840	0.8796	83.33%
	(A)	0.0965↑	0.3776↑	0.1531	0.2753↑	0.2753↑	0.3934↑	0.8807↑	73.81%

Table 6: Performance comparison category by each dimensions (B)efore and (A)fter neuron-level restrain.

Conclusion and Future Work

Contribution: RoleMRC is a comprehensive benchmark that pushes the boundaries of role-playing and instruction-following in LLMs.

Findings: Our experiments demonstrate the effectiveness of RoleMRC in training and evaluating more capable role-playing agents and reveal insights into the "alignment tax."

Future Directions:

- Incorporate more diverse system-level prompts.
- Address potential biases from synthetic data generation.
- Further investigate the broader impacts of neuron-level interventions.